| Region | Quadrant | Nature of x,y | Signs of the coordinates |
|---|---|---|---|
| $XOY$ | I | $x>0,\ y>0$ | (+,+) |
| $X'OY$ | II | $x<0,\ y>0$ | (–,+) |
| $X'OY'$ | III | $x<0,\ y<0$ | (–,–) |
| $XOY'$ | IV | $x>0,\ y<0$ | (+,–) |

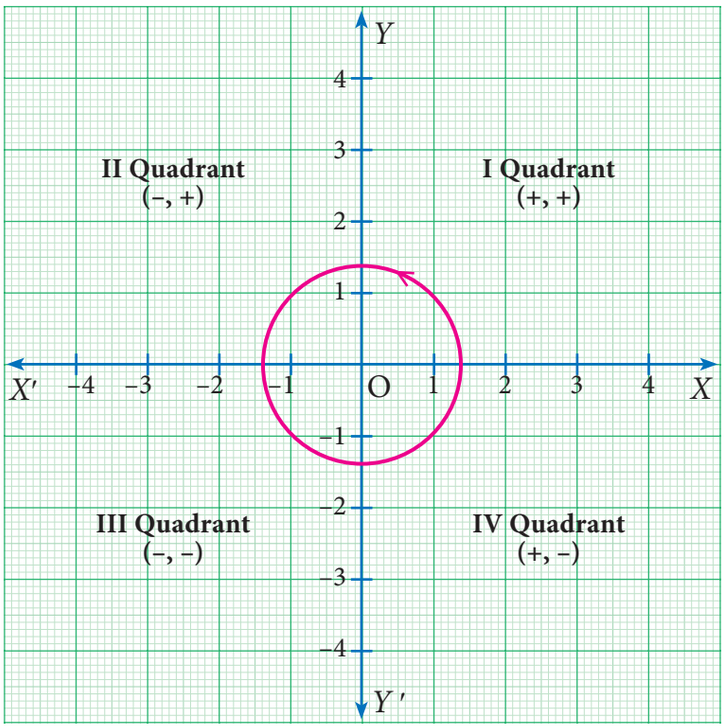



Fig. 5.7

**Note**

- For any point P on the x axis,the value of y coordinate (ordinate) is zero that is, P (x,0).
- For any point Q on the y axis, the value of x coordinate (abscissa) is zero. that is, Q (0, y)
- $(x, y) \neq (y, x)$ unless x= y
- A plane with the rectangular coordinate system is called the Cartesian plane.

### 5.2.1 Plotting Points in Cartesian Coordinate Plane

To plot the points (4, 5) in the Cartesian coordinate plane. We follow the $x$ – axis until we reach 4 and draw a vertical line at $x = 4$.

Similarly, we follow the $y$ – axis until we reach 5 and draw a horizontal line at $y = 5$.

The intersection of these two lines is the position of (4, 5) in the Cartesian plane.

This point is at a distance of 4 units from the $y$-axis and 5 units from the $x$-axis. Thus the position of (4, 5) is located in the Cartesian plane.

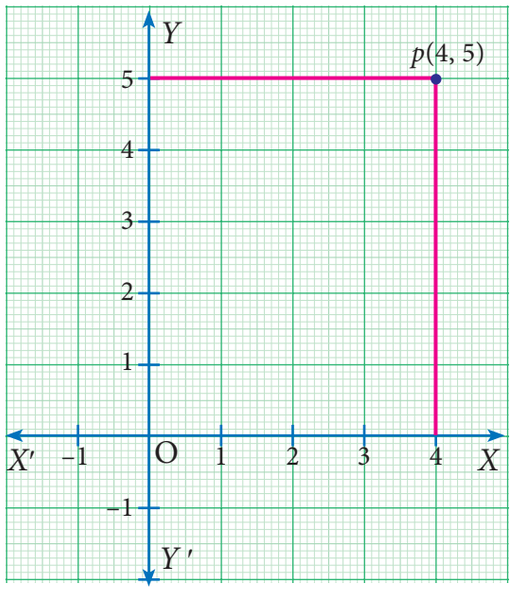



Fig. 5.8



5-Coordinat_1.indd 190 02/12/2022 21:44:42